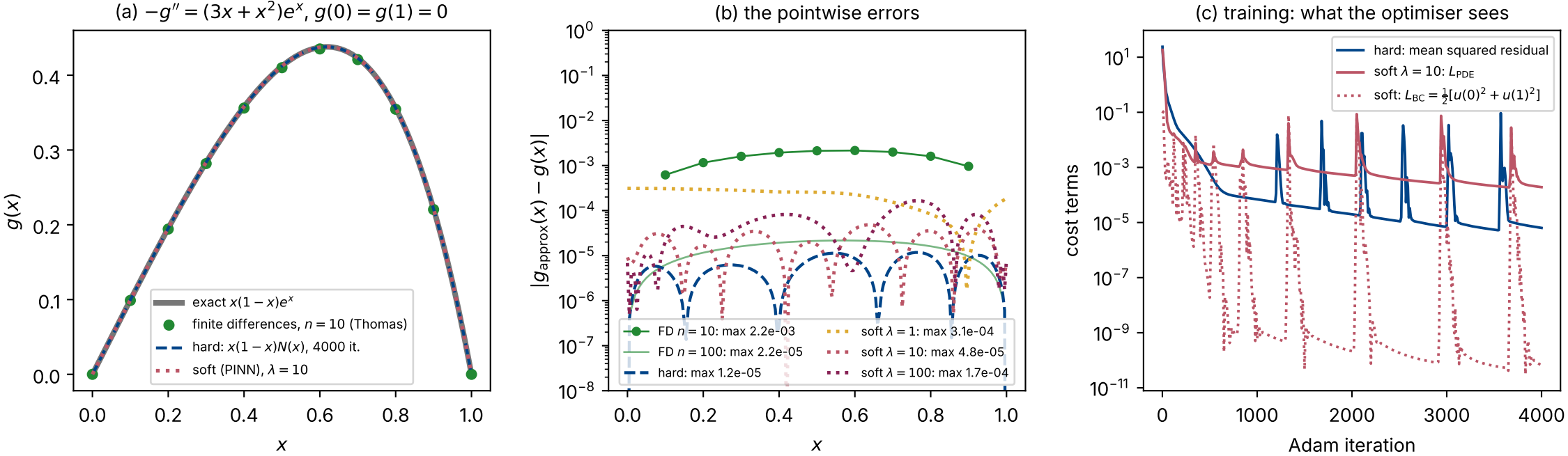
(a) $-g'' = (3x + x^2)e^x$, $g(0) = g(1) = 0$
$g(x)$
x
exact $x(1-x)e^x$
finite differences, $n = 10$ (Thomas)
hard: $x(1-x)N(x)$, 4000 it.
soft (PINN), $\lambda = 10$
(b) the pointwise errors
$|g_{\text{approx}}(x) - g(x)|$
x
FD $n = 10$: max 2.2e-03
FD $n = 100$: max 2.2e-05
hard: max 1.2e-05
soft $\lambda = 1$: max 3.1e-04
soft $\lambda = 10$: max 4.8e-05
soft $\lambda = 100$: max 1.7e-04
(c) training: what the optimiser sees
cost terms
Adam iteration
hard: mean squared residual
soft $\lambda = 10$: L_{PDE}
soft: $L_{\text{BC}} = \frac{1}{2}[u(0)^2 + u(1)^2]$